



Prediction of multi-wake problems using an improved Jensen wake model



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ABSTRACT

The improved analytical wake model named as 2D_k Jensen model (which was proposed to overcome some shortcomings in the classical Jensen wake model) is applied and validated in this work for wind turbine multi-wake predictions. Different from the original Jensen model, this newly developed 2D_k Jensen model uses a cosine shape instead of the top-hat shape for the velocity deficit in the wake, and the wake decay rate as a variable that is related to the ambient turbulence as well as the rotor generated turbulence. Coupled with four different multi-wake combination models, the 2D_k Jensen model is assessed through (1) simulating two wakes interaction under full wake and partial wake conditions and (2) predicting the power production in the Horns Rev wind farm for different wake sectors around two different wind directions. Through comparisons with field measurements, results from Large Eddy Simulations (LES) as well as results from other commercial codes, it is found that the predictions obtained with the 2D_k Jensen model exhibit good to excellent agreements with experimental and LES data.

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1. Introduction

Wind turbine wake effect, which is characterized by a significant deficit in wind speed and an increment of turbulence intensity, has been extensively studied in recent years. According to the work of [1] or [2], it was found that in large offshore wind farms, the average power loss due to wind turbine wakes is in the order of 10% (Middelgrunden wind farm) to 23% (Lillgrund wind farm) of the total energy production. For a wind energy project, accurate predictions of the wake flow behavior, such as the wind velocity distribution and the profile of turbulence intensity are quite important since the main work of the wind farm micro-siting is to minimize the impact of upstream wakes on the power productivity of downstream turbines.

The wake models which are commonly used to predict turbine wake flows can be grouped into analytical and computational ones. At current stage, full LES with resolved boundary layer for a single wind turbine wake is still too expensive. An alternative approach is

to use LES with Actuator Line (AL) method [3]. With the LES/AL method, it is possible to simulate multi-turbine wakes. However, these computational models are still too expensive as compared with the analytical models. So far they are not yet suitable for making wind farm parametric studies or layout optimizations. At present, simple analytical wake models are still the common tools to predict wind turbine wake effect. Especially in the case of wind farm layout optimization, the predictions of the wind turbines wake behavior need to be carried out for a large number of candidate farm layouts (necessary for optimization).

One of the most widely used and pioneering analytical wake models is the one proposed by Jensen [4] and later extended by Katic et al. [5]. The Jensen wake model (also named as Park model) has been extensively implemented in the literature [6,7] and the commercial software such as WASP [8], GH WindFarmer [9] and WindPro [10]. The main features in this model are that the wake expands linearly with downstream distance, and the velocity deficit has a top-hat shape in the wake regions. Through a set of numerical experiments [1,11,12], the Jensen wake model has been proved to provide an acceptable prediction on the power losses within wind farm.

However, the Jensen wake model has two major limitations. The first one is that the assumption of a top-hat distribution of the wake

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deficit is far from reality and cannot describe the velocity field accurately. The Jensen wake model provides a simple velocity profile within the wake field, as it assumes that the wake velocity only changes in the streamwise direction and keeps constant in the cross-wind direction. From the space perspective, this simple model is known as 1D (one-dimensional) wake model. However, the self-similar Gaussian or trigonometric shape of the velocity deficit has been observed in wind tunnel measurements [13], data of operating wind farms [14] and numerical simulations [2]. According to these observations, it has been concluded that the wake velocity actually varies in two-dimensional space and Gaussian or trigonometric distribution is more appropriate to describe the velocity deficit in the far wake region.

The second limitation is the assumption of a constant wake decay parameter through the entire wake flow field. The wake decay coefficient k describes how fast or slow the wake expands by specifying the wake growth rate as it travels downstream. It is recommended in the WAsP Help Facility [15] to use $k = 0.075$ for offshore cases and 0.04 or 0.05 for onshore ones. Later in the literature [16], a semi-empirical expression $k = 0.5/\ln(h/z_0)$ was proposed for deriving the wake decay constant, where h is turbine's hub height and z_0 is the terrain surface roughness height. But in references [17–19], the authors pointed out that this semi-empirical formula is not adequate because it is only related to the ambient turbulence levels. In fact, the growth of the wake is governed by many more factors, in addition to the ambient turbulence, there are shear-generated turbulence and the turbulence created by the turbine. Therefore, the wake decay rate should not be a constant but a variable parameter that relating to the effective wake turbulence which is composed of ambient turbulence and turbine-added turbulence.

An improved wake model was proposed in Ref. [20] based on the classical Jensen (Park) wake model and two modifications: (1) a cosine shape distribution instead of the top-hat shape distribution for velocity deficit profiles in the wake; (2) the wake decay parameter k (constant for the Jensen wake model) depending on both the ambient turbulence and the wake-generated turbulence, as well as the downstream distance from the wind turbine. The performance of this new model (named as 2D_k Jensen model) was extensively investigated for the prediction of a single wake in our previous work [20]. The results showed that the 2D_k Jensen model can give a more physical and accurate description of the velocity deficit in the single wake. The main contribution in the present paper is to apply and validate the 2D_k Jensen model for the prediction of multi-wake effects downstream of a series of wind turbines.

The paper is organized as follows: in section 2, a brief introduction of the 2D_k Jensen model is presented. The applications of the 2D_k Jensen model to predict wake interactions between two wind turbines (with different downstream and crosswind displacements) operating at different inflow conditions are given in section 3. After that, the proposed model is adopted to investigate the effect of multiple wakes on power losses in a large offshore wind farm in section 4. Finally, the summary and conclusions are presented in section 5.

2. Improved Jensen wake model for multiple wakes prediction

2.1. 2D_k Jensen wake model

In order to give the reader a clear picture, a brief summary of the 2D_k Jensen model [20] is made first. The Jensen wake model is regarded as one of the most useful analytical wake models, which has been implemented in the widely-used commercial software.

Based on the law of global momentum conservation, a mathematical expression is finally obtained to describe the wake velocity distribution downstream of a single wind turbine, which has the following form

$$u^* = u_0 \left[1 - \frac{1 - \sqrt{1 - C_T}}{(1 + kx/r_d)^2} \right] \quad (1)$$

where u^* is the wake velocity at a downwind position x , u_0 is the incoming wind speed, C_T is the wind turbine thrust coefficient, r_d is the rotor radius of the wind turbine, k is the semi-empirical wake decay constant which can be computed using

$$k = 0.5/\ln(h/z_0) \quad (2)$$

where h is the hub height of the wind turbine, z_0 is the surface roughness height of the local terrain. The Jensen model describes the wake to be linearly expanding, and the spreading wake radius r_x at the downwind distance x is formulated as

$$r_x = r_d + kx \quad (3)$$

This original Jensen wake model is referred to as the one-dimensional Jensen model as it considers the wake speed changing only with downstream distance (as shown in Eq. (1)). This model is also known as the top-hat model since the shape of the predicted wake velocity along the cross-stream direction is like a top-hat. However, according to the classical theories and numerical computations of shear flows in the wakes of bluff bodies [21] as well as the wind tunnel investigations of the wake behind a single wind turbine [13], the wake velocity varies both in downwind and crosswind directions and it has an approximately Gaussian, cosine or polynomial shape in the crosswind direction after a certain downstream distance. Therefore, the Jensen wake model is too simple to describe the realistic distribution of wake deficit in the crosswind plane, a modification is highly needed to represent the physical wake velocity distribution.

In our previous work [20], the 2D_k Jensen model was proposed, in which the velocity profile in the cross section was represented by a cosine shape instead of a top-hat shape in the standard Jensen wake model. The expression of the developed 2D_k Jensen model is given as follows

$$u = (u_0 - u^*) \cos\left(\frac{\pi}{r_x} \cdot r + \pi\right) + u^* \quad (4)$$

where u^* is calculated using Eq. (1). It is worth noting that the proposed 2D_k Jensen model consists of two steps: the prediction of the original wake velocity at certain downwind position using the Jensen wake model; and then the redistribution of the calculated wake velocity along the crosswind direction using the cosine shape function.

For using the Jensen wake model, a crucial problem is to define the wake decay parameter k . The k value represents how the wake breaks down by specifying the growth of the wake width per unit length in the downwind direction. In the original Jensen model, the parameter k is supposed to be a constant with the value obtained from the semi-empirical function, Eq. (2) or some empirical instructions [8]. But researches in references [17–19] all showed that this approach is too realistic and not consistent with the ambient turbulence level, in which the roughness height is considered to be the only one influence factor on the wake decay parameter k . In fact, the wake growth is governed by the atmospheric turbulence as well as the turbulence generated by the rotor itself, this implies that instead of using a constant, the wake decay parameter k can be

changeable take into account the a variable turbulence level throughout the entire wake field.

The turbulence intensity determines the level of turbulence mixing and subsequently determines the wind deficit recovery rate and the wake width. Based on this fact, a modified wake decay parameter was proposed in our previous work [20], which is assumed to have the following form

$$k = k_0 \frac{I_{wake}}{I_0} \quad (5)$$

where I_0 is the ambient turbulence intensity, I_{wake} is the effective turbulence intensity in the wake flow area and k_0 is the wake decay constant given by the Jensen model. This equation takes into account the effect of effective turbulence on the wake recovery, and states an increased turbulence level resulting in a wider wake width and a faster wake recovery.

In Ref. [22] the authors validated four widely-used, state of the art, analytical turbulence models, including the Quarton model, the Larsen model, the Frandsen model and the model recommended by IEC 61400. These models indicate that the total effective turbulence within the wake is assumed to be covered by the atmospheric turbulence intensity I_0 and the turbine added turbulence I_{add} , and this gives rise to an expression for the total turbulence intensity along the wake centerline,

$$I_{wake} = \sqrt{I_0^2 + I_{add}^2} \quad (6)$$

Through a series of numerical tests, it is found that the Larsen model [23] most accurately predicts the total turbulence intensity, with the expression for the turbine added turbulence (for downstream spacing larger than 2D) as below

$$I_{add} = 0.29 \left(\frac{x}{D}\right)^{-\frac{1}{3}} \sqrt{1 - \sqrt{1 - C_T}} \quad (7)$$

Same as in Eq. (1), x denotes the downwind distance from the wind turbine. Inserting Eq. (7) into Eq. (6), one can obtain the total turbulence intensity at any downstream position in the wake area. Assuming that the initial value of turbulence intensity is $I_0 = 8\%$ and the thrust coefficient of the wind turbine C_T is 0.78, then the downstream development of dimensionless wake turbulence intensity obtained by Larsen model is shown in Fig. 1. Note that the prediction is started from the position 2D downstream of the rotor.

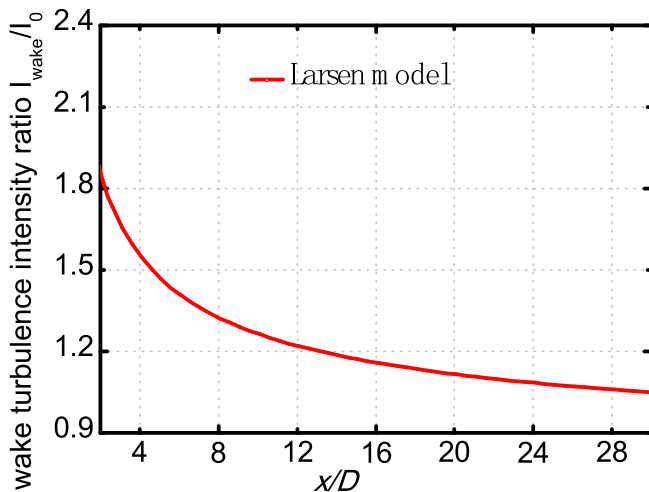


Fig. 1. Distribution of wake turbulence intensity calculated using Larsen model.

In this figure an obvious increased turbulence level is clearly seen in the near wake region ($x < 5D$) and then the turbine added turbulence gradually decrease as the distance from the turbine rotor increases, finally the wake turbulence recovers to the incoming level in the further downstream positions.

To sum up, the proposed new wake model was based on the two above mentioned modifications: the first one is that the top-hat distribution is corrected to a cosine distribution for the velocity deficit in the wake, which makes the velocity distribution two-dimensional; another one is that the effect of turbulence on the wake recovery from both the ambient turbulence and the turbine generated turbulence is taken into account, resulting in a variable wake decay parameter k (changing with the effective turbulence in the wake) instead of a constant one given by the Jensen model. On the basis of these corrections, the 2D_k Jensen model was developed with the letter “2D” and “k” represents the two-dimensional feature and the corrected parameter k of the new model, respectively. The equations for the 2D_k Jensen model are given by:

$$u^* = u_0 \left[1 - \frac{1 - \sqrt{1 - C_T}}{\left(1 + \frac{kx}{r_r}\right)^2} \right] \quad \text{with} \quad k = k_0 \frac{I_{wake}}{I_0} \quad (8)$$

$$u = (u_0 - u^*) \cos\left(\frac{\pi}{r_x} \cdot r + \pi\right) + u^*$$

2.2. Multiple wakes combination effect

The wake model discussed in the previous section only considers the wake effect behind a single wind turbine. However, in a wind farm, many wind turbines might be affected by dozens of upstream turbines, i.e. the wind speed at one turbine is affected not only by the wind turbine that is directly in front of it, but also by other upstream wind turbines. In this case, the multiple wakes must be combined. There are four common wakes combination models that are used to simulate the interaction effect between these multiple wakes [24], from which the effective wind speed in the wakes of multiple turbines can be calculated

$$\begin{aligned} \text{Geometric sum (GS)} : \quad & \frac{u}{u_0} = \prod_{j=1}^n \frac{u_j}{u_0} \\ \text{Linear superposition (LS)} : \quad & \left(1 - \frac{u}{u_0}\right) = \sum_{j=1}^n \left(1 - \frac{u_j}{u_0}\right) \\ \text{Energy of balance (EOB)} : \quad & u_0^2 - u^2 = \sum_{j=1}^n (u_0^2 - u_j^2) \\ \text{Root sum of squares (RSS)} : \quad & \left(1 - \frac{u}{u_0}\right)^2 = \sum_{j=1}^n \left(1 - \frac{u_j}{u_0}\right)^2 \end{aligned} \quad (9)$$

where u is the wind speed at any research positions in the wake area, u_j represents the wake speed at the downstream place due to the single wake from upstream turbine j (which can be obtained from Eq. (8)) and n is the number of upstream turbines.

In order to explain the multiple wakes combination process we consider an example of wind farm with known coordinates for each wind turbine location within the farm. Based on this, the multiple wakes cumulative process can be divided into the following steps: (1) the incoming wind speed u_0 is evaluated starting with an undisturbed turbine positioned at the most upstream position; (2) using Eq. (8), calculate the velocity deficit downstream of this

turbine for all downwind turbine locations progressively; (3) continue with the next second upstream turbine, even though it is positioned in the wake area of the first upstream turbine, still employing the free stream wind speed u_0 instead of its real incoming wind speed, and then calculate the wind speed downstream of it for all downstream turbine positions; (4) repeat the calculation in the same way for the rest of turbines, and then all wake deficit from the upstream turbines that affect a downwind candidate turbine can be obtained; (5) using Eq. (9), combining all single wakes into a resultant wind speed u for the candidate turbine according to the afore mentioned wakes combination laws. It needs to be pointed out here that since the single wake profile computed from the improved Jensen model has a two-dimensional feature, the resultant wake profile obtained through cumulating a number of single wakes will has a two-dimensional feature accordingly.

All four combination models are incorporated into the commercial software GH WindFarmer as the main algorithms to consider the multiple wakes effect. The performances of these combined models were tested and compared with the wind tunnel data [24] and it was found that the root sum of squares model is the best performing method for almost all cases, followed by the energy balance model. Furthermore, the methods of geometric sum and linear superposition are not recommended since they tend to overestimate the velocity deficit. In the software WindPRO, only the root sum of squares model is available. Different from these two software, the geometric sum model (which is not recommended in GH WindFarmer) is suggested in ECN's software tool FluxFarm. The accuracy uncertainties of these models demonstrate the needs for more comprehensive validations of them. In our following sections, the performances of these models will be further tested and the differences between them will be presented and discussed.

2.3. Partial wake effect

In a real wind farm, wind comes from any directions, this makes the wind not always aligned with the columns of wind turbines. Therefore, the wake shadow area of the upstream turbine does not cover downstream turbines all the time. The situation depends on the properties of the wind farm such as the total number of turbines, the layout and the separation distance between individual turbines. When the uncertainty of wind direction is considered, upstream wakes can affect fully or partially over downstream turbines. Since the model discussed in the previous section is built for full shadowing, it is fundamental to consider the partial wake effect within wind farm.

It is needed to state out the difference between the way of the original Jensen model and the 2D_k Jensen model in considering the partial wake effect. For a certain downstream turbine, if only one part of its rotor plane is touched by the wake of the upstream one, the wake velocity deficit will be over-predicted when the original Jensen model is directly used. At this time, partial wake correction that considering the effective percentage for the overlapping between the area of the wake produced by upstream turbine and the rotor swept area of downwind turbine is highly needed for such cases. But for the 2D_k Jensen model, since the model has a two-dimensional feature, there is no need to introduce any partial wake correction into the model.

3. Study of two wind turbine wakes interaction

3.1. Single wind turbine wake test

Before the study of two wind turbine wakes interaction, the investigation of the wake flow downstream of a single wind turbine is performed at first. The performance of the 2D_k Jensen wake

model in predicting single wind turbine wakes is assessed through comparisons with field measurements in a wind farm located at the Tjæreborg Enge site. This wind farm is composed of eight turbines of 2 MW NM80 and V80, which have a diameter and a hub height of 80 m and 75 m, respectively. A meteorological mast is installed to provide the single wake measurements at 3.9D downstream of the 2 MW NM80 wind turbine. All the data used here are extracted from the Database of Wind Characteristics with a reference time period of 10 min [25]. Because the free wind measurements was not available, the incoming wind speed was approximated from the measured electrical power output using the power curve of the upstream turbine, while the ambient turbulence level was estimated from the level measured by a met mast for wind directions just outside of the wake area. Based on the analysis of meteorological data, the free stream wind speed was approximately 8 m/s, while the ambient turbulence intensity at hub height was 10%. Under this condition, the thrust coefficient was estimated to be 0.806.

Fig. 2 shows the horizontal profile of the normalized velocity deficit and turbulence intensity at turbine hub height as a function of normalized axial distance for different analytical models. The field measurements and the LES data were presented in Ref. [25]. To fully reflect the scatter of the measurements, error bars which represent the standard deviation of the measured 10-min averages are also plotted in the figure. From the left figure it can be seen that the top-hat assumption makes the Jensen wake model not only to under-predict the velocity deficit at the wake center area but also to overestimate it near the edge of the wake. On the other hand, the results obtained with the 2D_k Jensen model are in acceptable agreement with the field measurements as well as the AL/LES simulation results (the simulation was performed employing the actuator line technique combined with large eddy simulation method, and the incoming wind condition for the calculation, such as wind speed and turbulence intensity, is the same as in the present work), except that measurements show a non-axisymmetric velocity distribution in the wake and the AL/LES model can partly capture this asymmetric behavior but the 2D_k Jensen model exhibits a cosine axisymmetric shape. This discrepancy might be due to differences in the inflow and operational conditions, but also due to the limited amount of available measured data.

As seen in the figure, the turbulence level prediction model (the 2D_k Jensen model) reproduces the turbulence intensity of the wake flow rather well. The predicted turbulence intensity is closer to the mean value of measured 'double-peaked' structure (these double peaks are located near the position of maximum gradient in the velocity profiles and thus represents the maximum turbulence level in the shear-layer). This is in good agreement with the measured turbulence intensity. The improved model proves a better accuracy of the Larsen turbulence model to some extent. It is noticed that the Larsen model can provide an approximately maximum value of the turbulence intensity as a reference for the wind turbine design process, the maximum turbulence intensity is an important parameter for the design since the wake turbulence of the upstream wind turbine has a strong impact on the fatigue loads as well as the lifetime of the downstream one.

3.2. Two wind turbines wakes interaction test

For wind farm applications, it becomes more challenging for analytical wake models. The analytical models should not only predict the wake effect of a single wind turbine correctly, but also the interaction of multiple wakes as well. Besides, it is common in most offshore wind farms that the turbines are arranged in-line, but the downstream wind turbines are slightly offset so that the wakes of upstream wind turbines only affect partially of the area swept by

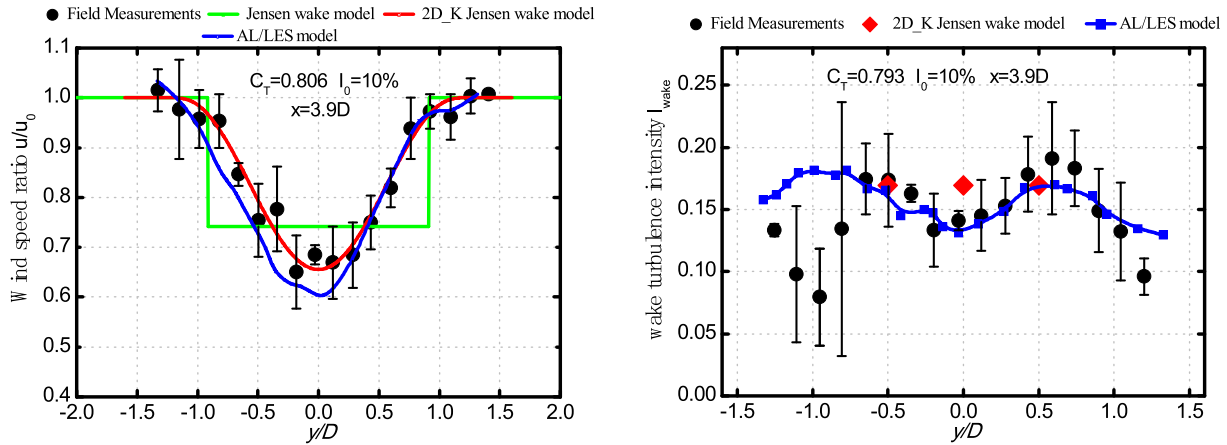


Fig. 2. Comparisons of the predicted and measured mean velocity (left) and turbulence intensity (right) along the crosswind direction at hub height and a downstream position $x = 3.9D$.

the downstream turbine, which means the rotor of the downstream wind turbine experiencing both the undisturbed and wake affected flow. In this case, the produced flow fields are complicated which causes more damage to the downstream turbines. To gain insight in the capabilities of the proposed 2D_k Jensen wake model (coupled with four wake combination models outlined in section 2.3) in predicting the velocity deficit throughout the wakes induced by two wind turbines with varying mutual distances, simulations employing the 2D_k Jensen wake model are carried out with different downstream and crosswind displacements between the rotors. It should be mentioned that just like the employed incoming wind condition for the single wake case, in this case the mean wind speed was assumed to be 8 m/s, while the turbulence intensity at hub height was estimated to 10%.

Fig. 3 shows the development of the stream-wise velocity behind the two wind turbines with the downstream turbine (denoted turbine T2) where T2 is shifted 3.3D downwind of the upstream one (denoted as turbine T1), and the configuration of the turbine is indicated by bold vertical black lines. Two horizontal profiles are extracted at specific stream-wise positions, one close to turbine T2 at $x = 7.0D$ and the other one further downstream at

$x = 9.0D$ (the coordinate is defined to be relative to the first turbine T1). Comparing the wake deficit profiles at $x = 7.0D$ and $x = 9.0D$, it can be observed that the 2D_k Jensen wake model coupled with RSS (root sum of squares) combination model shows the best prediction of the wake width (i.e., the horizontal distance from the wake center-line to the position where 99% of the free-stream velocity has been recovered) and wake depth (i.e., the maximum velocity deficit at the wake center-line). The 2D_k Jensen model coupled with EOB (energy of balance) comes the second with a good estimation of the wake depth but an under-prediction of the wake velocity is shown in the crosswind positions. The methods geometric sum (GS) and linear superposition (LS) tend to overestimate the velocity deficit, especially the wake depth is overestimated up to 20% as compared to 0.5% for the RSS model at the position $x = 7.0D$, and this overestimation is more obvious as it goes further downstream, e.g. $x = 9.0D$.

When a variable wind direction is considered, upstream wakes will affect partially or totally over downstream turbines. Generally, 0° refers to the row direction; if the wind direction deviates from 0° , it is practical to retain the main flow direction parallel to the x -axis and change the positions of the downstream wind turbines

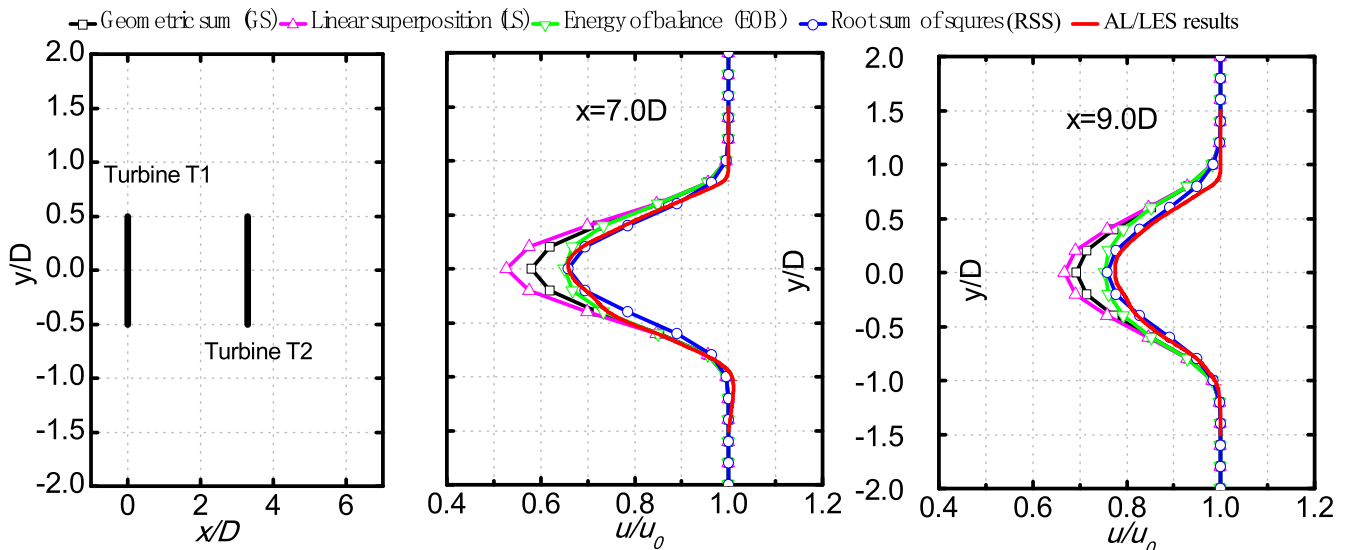


Fig. 3. Profiles of velocity deficit in a horizontal plane at hub height for two wind turbines separated with stream-wise and cross-wind distances of 3.3D and 0.0D, respectively.

because their orientation is always parallel to the wind direction. In order to study the wake interaction between two wind turbines operating at different inflow conditions, simulations are carried out for the case that the second turbine T2 shifted 0.6D in the crosswind direction off the center-line $y = 0$ (which can represent the deviation angle about 10°), as shown in Fig. 4.

Using the 2D_k Jensen model, it can be shown that the wake radius after 3.3D downstream of the first turbine T1 is about 0.84D, so that the wake of turbine T1 does not fully cover the rotor area of the downstream turbine T2, in which the partial wake effect has to be incorporated into the superposition model. In Fig. 4, the velocity deficits obtained with the 2D_k Jensen model coupled with four multiple wake combination models are compared with the simulation results of the AL/LES model. Similarly to the results presented in Fig. 3, it can be seen that the 2D_k Jensen model coupled with the RSS model exhibits the best agreement with the AL/LES simulation results, and then followed by the EOB model while the GS model comes the third, and the LS model appears to lose much accuracy in this case.

From Fig. 4 it can be observed that the profiles of the wake velocity consist of three regions, the region ($0.25 < y/D < 1.5$) where the wake is only affected by upstream turbine T1, the central part of the wake ($-1.25 < y/D < 0.25$) is a mixture of influences from the two wakes, and in the region ($-2.0 < y/D < -1.25$) only the effect of downstream turbine T2 is observed. Compared to the wake behind a single wind turbine as illustrated in Fig. 2, we can observe that, the maximum velocity deficit appears at the center of the interacting wake, which tends to move toward to the central axis of the downstream turbine T2. As expected, this observation indicates that the development and the shape of the mixing wake is mainly driven by the nearest upstream turbine. When compared to the mixing wake behind the two wind turbines displaced in a line as shown in Fig. 3, it can be found that the depth of the wake deficit at the wake center (where a strong wake interaction occurs) for the full wake case is a little bigger than the depth of the wake deficit for this partial wake case, which can be explained by the fact that the center of the blending wake experiencing much strong wake deficit from the first turbine T1 for the full wake situation than the partial one.

Numerical investigations are also conducted on the above two wind turbines with turbine T2 moved further downstream of turbine T1, which makes the stream-wise separation between the two

rotors to be 6.6D. The crosswind velocity profiles at 10.0D downstream of turbine T1 are plotted in Fig. 5, in which the results obtained by the 2D_k Jensen model coupled with four wakes combination models are compared with predictions employing AL/LES method [25]. It shows that the 2D_k Jensen model in combination with the RSS model leads to the best agreement with the LES data and is followed closely by the EOB model, while the other two wakes combination models tend to over-predict the wake deficit by up to 15%. Similar trend has been observed as in the former case. These comparisons again prove the good accuracy of the proposed 2D_k Jensen model and in the meanwhile show the applicability of the RSS wakes combination model.

When comparing the results in Fig. 5 with those in Fig. 3, it could be found that the crosswind velocity profiles at a downstream position $x = 10.0D$ in the former plot and the velocity profiles at a downstream location $x = 7.0D$ in the latter one are close to each other. More specifically, it is found that the minimum wind speed ratio at the wake center is measured to be 0.66 and the wake radius is 1.1D at the downstream position $x = 10.0D$ in Fig. 5, which are almost consistent with the results in Fig. 3 that the minimum centerline velocity and the wake radius are 0.65 and 1.03D, respectively. These comparisons indicate that the properties of the wake at these two positions are quite similar in the aspect of centerline wake deficit, wake width and shape of the wake profile. Since the position $x = 10.0D$ is located at 3.4D far from turbine T2 as seen in Fig. 5 while the location $x = 7.0D$ is 3.3D downstream of turbine T2. This again supports the finding that the development and the shape of the mixing wake is mainly governed by the nearest upstream turbine, also meaning that the resulting wake effect in the multiple wakes seem to be equal to the maximum effect of all upstream turbines.

Additionally, the influence of the deviation angle on the wake interaction effect is also studied with a lateral spacing of 0.75D for the two turbines positioned at 6.6D from each other in the stream-wise direction. Fig. 6 shows the wake speed at the position of $x = 10.0D$ downstream from turbine T1 (thus 3.4D downstream from turbine T2) predicted by the 2D_k Jensen model coupled with four different wakes combination models, and compared against the AL/LES results reported in Ref. [25]. It is observed that the 2D_k Jensen model combined with the RSS model can predict the velocity deficit reasonably well in the regions of $y/D < -0.5D$ and $y/D > 0.0D$, especially the predicted maximum wake deficit in the

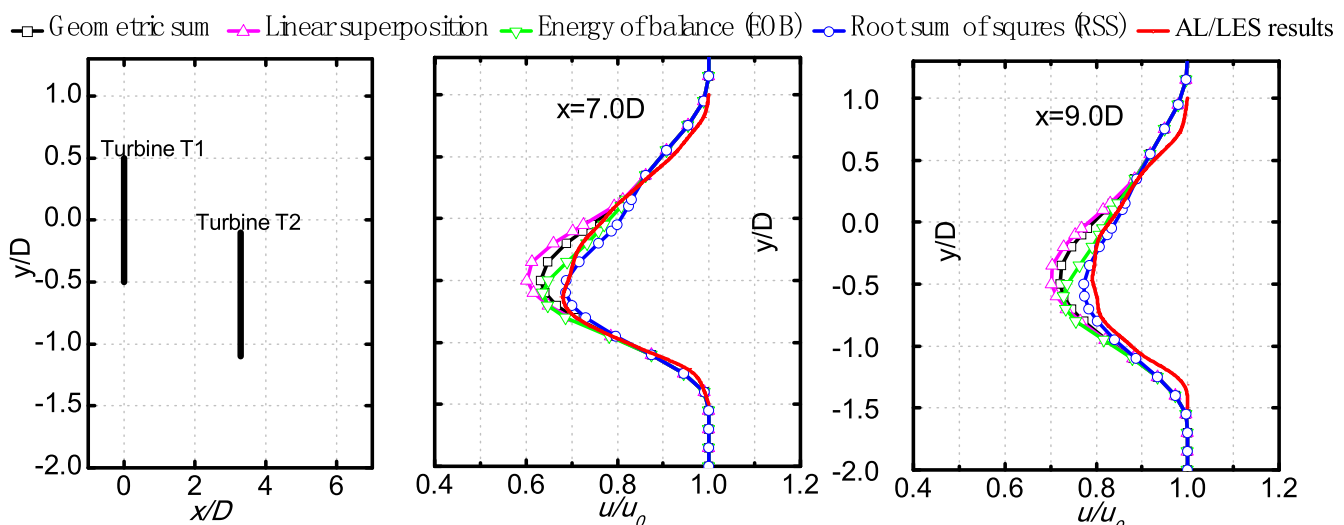


Fig. 4. Profiles of the velocity deficit in a horizontal plane at hub height for two wind turbines separated with stream-wise and cross-wind distances of 3.3D and 0.6D, respectively.

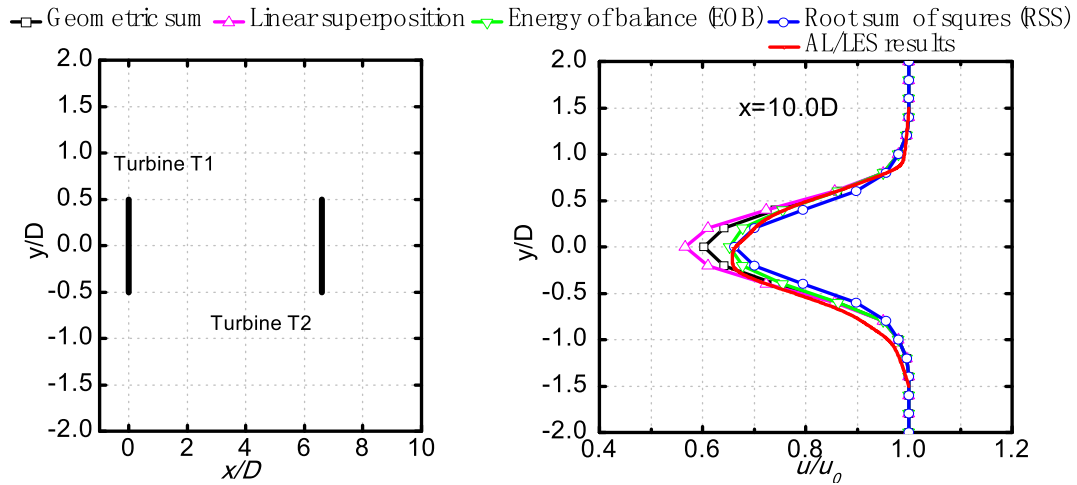


Fig. 5. Profiles of velocity deficit in a horizontal plane at hub height for two wind turbines separated with stream-wise and cross-wind distances of 6.6D and 0.0D, respectively.

mixing wake center is in fair agreement with the LES data, but this model underestimates the velocity deficit in the small region of $-0.5D < y/D < 0.0D$ with respect to the LES data. While the other three wakes combination models produce very similar results, they tend to over-predict the wake deficit in the mixing wake area, with a greater discrepancy in the region between the axis center of turbines T1 and T2.

As compared with the profiles of the wake deficit at the same downstream position $x = 10.0D$, In Figs. 5 and 6, it is seen that for the full wake condition the wake velocity has a symmetric distribution in the crosswind direction and the center of the mixing wake (where the maximum velocity deficit is found) is positioned along the axis line (also the wake center line) of the two turbines; while for the partial wake case the wake velocity presents a non-axisymmetric distribution and the center of the merging wake occurs at a lateral position between the rotor centers of the two turbines, very close to the extended line of turbine T2's axis. This well proves the conclusion that the position of the wake center in the merging wake is mainly determined by the nearest upstream turbine. Besides, it is worth mentioning that the 2D_k Jensen model joined with the RSS combination model can thoroughly capture the non-axisymmetric feature of the velocity distribution due to the presence of a certain deviation angle, which reveals the good

performance of our proposed wake model and also the desirable quality of the RSS wakes combination model.

4. Power prediction in a large offshore wind farm

Developing profitable wind energy projects requires a robust wind power assessment campaign and one of the key aspects of an energy analysis is the prediction of the energy losses due to wind turbine wakes. Besides that, the impacts of multiple wakes are also of critical importance to the wind energy industry because they directly impact the turbulence level that determines turbines' lifetime. To increase the understanding of wake effects in large wind farms, a number of field measurements have been carried out. Among others, the Horns Rev offshore wind farm [14,26,27] is selected as a case study, since it has been extensively studied with a plenty of meteorological and SCADA data which are available for comparisons. Furthermore, this wind farm is quite representative where the turbines are arranged in a regular geometrical shape and the separation distance between neighboring turbines is 7D (Normally, in the wind energy project, a rule of thumb mentions that the suitable separation distance between two adjacent turbines is 7–15 rotor diameters in the predominant wind direction [28]).

The Horns Rev wind farm is located in the North Sea 14 km off

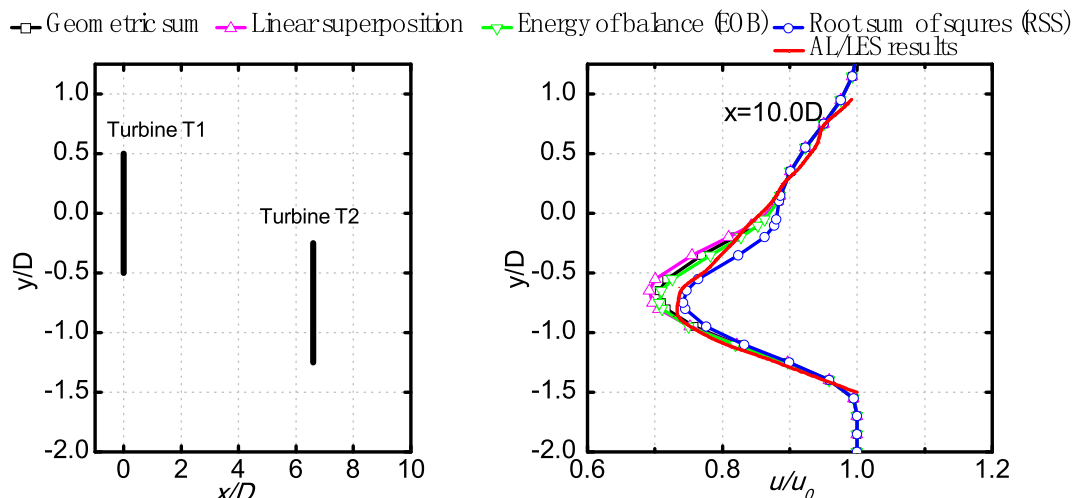


Fig. 6. Profiles of velocity deficit in a horizontal plane at hub height for two wind turbines separated with stream-wise and cross-wind distances of 6.6D and 0.75D, respectively.

the west coast of Denmark, with a water depth of 6–14 m. This wind farm consists of 80 Vestas-V80 2 MW wind turbines arranged in a matrix shape of 10 rows (west to east) by 8 columns (north to south) with the turbine spacing distance in both rows and columns is 7D, as shown in Fig. 7. The vertical rows are aligned approximately 7.2° turned from the north-south direction, forming a rhomboid shape. From the layout of wind turbines, it can be seen that the separation distance between two consecutive turbines highly relies on the incoming wind direction. For instance, the internal spacing along the main direction 270° is 7.0D and the spacing becomes 9.4D when the wind direction changes into 221°, as the solid lines with arrows in Fig. 7. These two situations are chosen as the numerical test cases to evaluate the performance of the proposed 2D_k Jensen wake model when predicting full wake condition. In order to investigate the full wake and partial wake effect, different direction intervals such as $[-1^\circ, 1^\circ]$ and $[-5^\circ, 5^\circ]$ are considered.

The Vestas-V80 wind turbine is a 2 MW, pitch controlled, variable speed wind turbine with rotor diameter of $D = 80$ m and hub height of $H = 70$ m (above sea level). Manufacturer's power curve and thrust coefficient as functions of wind speed are shown in Fig. 8. Using the combined rotor speed and pitch control one can obtain the constant thrust coefficient of 0.8 for the wind speed range 4–10 m/s as indicated in this figure. The Horns Rev wind farm is characterized by low turbulence level ($<8\%$) and many operational hours in near-neutral stability state. The Horns Rev data set contains an amount of representative 10-min statistical electrical power values from each wind turbine and the wind speed used for comparison in the following part is derived from the measured power along with the power curve of Vestas V80 wind turbines.

4.1. Single wind turbine wake test

Before carrying out the study of multiple wakes, the simulation of a single wind turbine wake is performed to verify the accuracy and generality of the 2D_k Jensen model. Based on the numerical investigations of V80 wind turbines reported by Wu et al. [2], the turbine is placed on a flat surface with a roughness length $z_0 \approx 0.005$ m and the incident wind speed at hub height is about 9 m/s, while the turbulence intensity at hub height is around 7%. Under this condition, the thrust coefficient is $C_T \approx 0.8$. Horizontal profiles of the velocity deficit at two representative positions $x = 7.0D$ and $x = 10.0D$ are shown in Fig. 9, in which the predictions obtained with the original Jensen model and the 2D_k Jensen

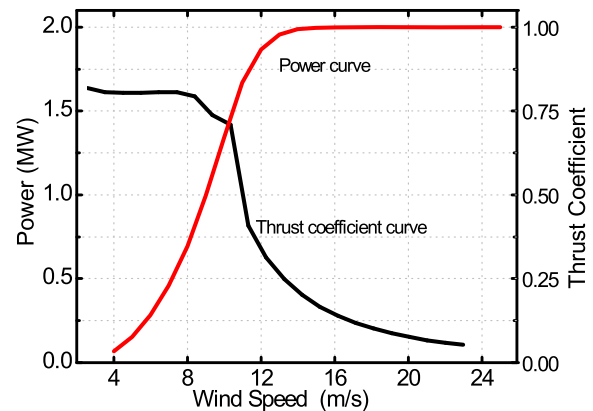


Fig. 8. Power curve and thrust coefficient curve of Vestas V80 2 MW wind turbine.

model are present. Besides, the numerical results calculated using a LES model [2] coupled with an actuator disc method are included in this figure for comparison.

It is shown that the Jensen wake model overestimates the wake velocity in the wake center area and underestimates it near the edge of the wake at these two downstream positions. For the performance of the 2D_k Jensen model, it is clearly seen that this model slightly under-predicts the wake deficit, relatively to the LES simulations, at a downstream position of $x = 7.0D$. As the downstream distance increases to $x = 10.0D$, the difference between the 2D_k Jensen model predictions and the LES data decreases, and an excellent agreement is obtained at this position. It should be mentioned here that we use LES results for comparison instead of measurements, from this point of view it seems that this test case cannot convincingly provide the accuracy of the 2D_k Jensen model, but at least these comparisons show that the predictions of the developed 2D_k Jensen model compare reasonably well with the results of an advanced CFD model in the far wake region.

4.2. Multi-wake tests (wind direction 221°)

The preliminary evaluation of the 2D_k Jensen wake model performance when predicting multiple wake effect in Horns Rev is for a south-westerly wind direction with flow exactly along the columns (221°) is shown in Fig. 7. The reference wind speed calculated from the power output of the first wind turbine is 8 ± 0.5 m/s, and the separation distance between consecutive turbines is 9.4D. Fig. 10 displays the normalized wind speed deficit and power output for the wind direction of 221° with a narrow sector $[-1^\circ, +1^\circ]$ as a function of the turbine row number, in which the numerical results are obtained using the 2D_k Jensen model coupled with four multiple wakes combination models. The error bar included in this plot denotes the standard deviation of the measured 10-min averages and thus represents the scatter of the measurements.

About the effect of the developed wake model and the multiple wakes combinations models, the comparisons in Fig. 10 reveal that the 2D_k Jensen model coupled with the RSS model matches almost perfectly with all the measured points, and then followed by the EOB combination model and the GS model while the linear superposition model performs worst with the wake velocity and their related power output is under-predicted by up to 9% and 24%, respectively. It needs to be pointed out that the numerical simulations are performed under the situation that the deviation angle is 0° (which refer to the coming inflow is parallel to a straight line connecting the hubs of a series of turbines), whereas the measured

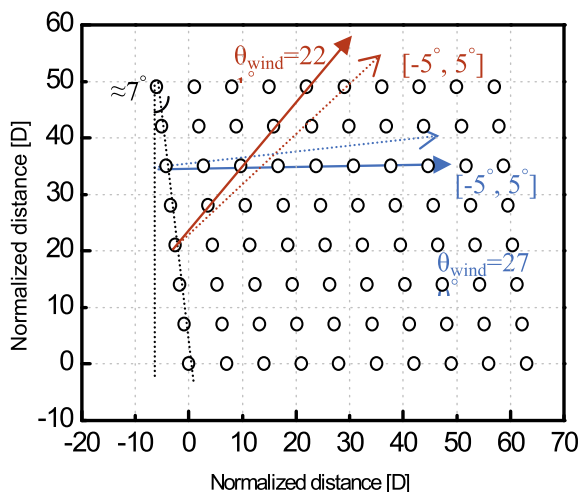


Fig. 7. Layout of the Horns Rev wind farm.

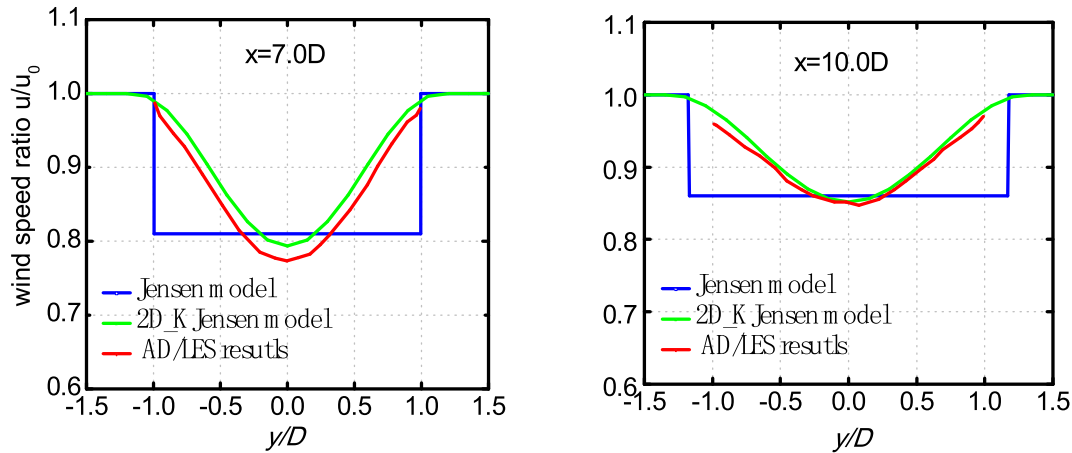


Fig. 9. Wind speed ratio in the cross wind direction at downstream positions of 7.0D (left) and 10.0D (right).

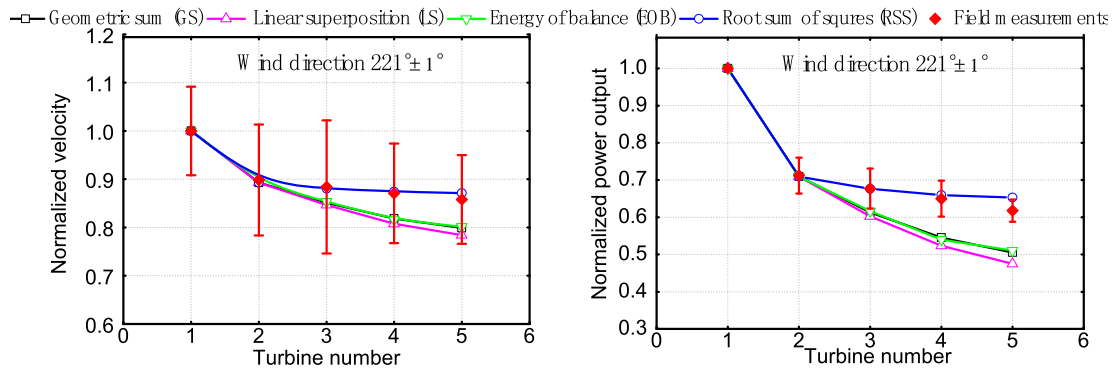


Fig. 10. Comparisons of the measured (with an interval of $[-1^\circ, +1^\circ]$) and the simulated (the deviation angle is 0°) normalized wake velocity (left) and normalized power production (right) for the wind sector of $221^\circ \pm 1^\circ$.

data are collected for a narrow sector of $\pm 1^\circ$. The small sector size of 2° represents an almost pure wake situation where all downwind turbines are covered by the wake, so from this point of view the simulated and measured data correspond to full wake situation. The coincidence between the numerical predictions and the field measurements are seen in Fig. 10. The results show a good quality of the 2D_k Jensen model when predicting the full wake situation.

When the sector size is increased to 10° , the comparisons of the simulated wake deficit as well as the power production with measurements for each turbine are shown in Fig. 11. In this plot, the good performance of the 2D_k Jensen model in combination with

the RSS model is clearly shown, where the predicted power output has been improved up to 20% as compared with the other three multiple wakes combination models. Although the 2D_k Jensen model coupled with RSS model performs better than the others, a small discrepancy between the model and the observation about wake velocity and, in turn, power output still exists. This is mainly due to the fact that the numerical results in Fig. 11 are obtained under the situation that the deviation angle is 2.5° , which is the intermediate value of the wider direction interval $[-5^\circ, +5^\circ]$. By contrast, the measurements are collected from the statistical analysis of the observed data relating to the bin $\pm 5^\circ$.

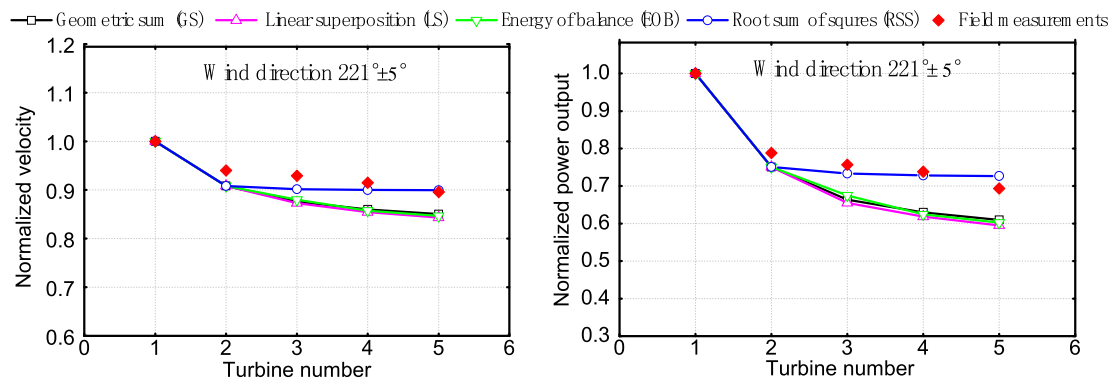


Fig. 11. Comparisons of the measured (with an interval of $[-5^\circ, +5^\circ]$) and the simulated (the deviation angle is 2.5°) normalized wake velocity (left) and normalized power production (right) for the wind sector $221^\circ \pm 5^\circ$.

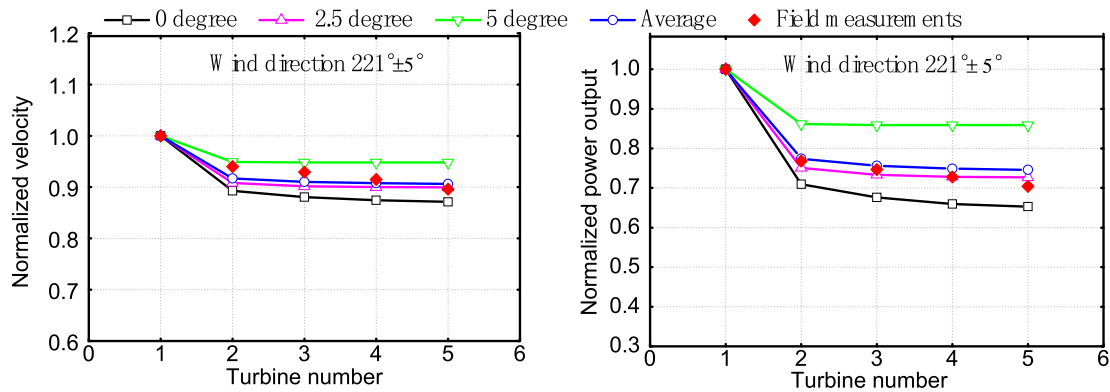


Fig. 12. Simulated wake deficit and power production using the 2D_k Jensen model coupled with the RSS combination model for three deviation angles: 0°, 2.5° and 5°.

In order to fairly compare the numerical results with measurements, simulations using the 2D_k Jensen model combined with the RSS model for a deviation angle of 5° is carried out and the results are presented in Fig. 12. In addition to that, the predictions corresponding to the deviation angles of 0° and 2.5° are also included in this figure. Here we average the simulated results obtained from the three cases (0°, 2.5° and 5°) and finally get the mean value as shown in Fig. 12. It can be noticed that there is a certain discrepancy between the averaged numerical results and measurements for the downstream turbines. As for the velocity deficit, the measured data shows that the wind speed drops at the second turbine to about 94% and keeps decreasing along the line of turbines, while the averaged predictions reveal that the wind speed declines to about 92% but then remains approximately constant further downwind. As expected, the trend for the associated power production is nearly the same. Furthermore, it is interesting to find that the wake velocity is slightly under-predicted by the average method but the power output is over-predicted a little bit as compared with the measurements. This puts forward a question that the resulting power output should be predicted by averaging the output for three cases (0°, 2.5° and 5°) or derived by the averaged velocity through the known power curve of the Vestas-V80 wind turbine. Although there is a small difference behind the recovery gap, in general, the agreement between computed and measured data is good.

4.3. Multi-wake tests (wind direction 270°)

The flow direction at 270° is a column consists of 10 wind turbines with 7D spacing, but only 8 upstream turbines in each

column have been investigated in the wake deficit measurement project. The comparisons of normalized wake wind speed and normalized turbine power data obtained from the developed 2D_k Jensen model combined with the four different multiple wakes combination models for a wind sector of $\pm 1^\circ$ around the wind direction of 270° are illustrated in Fig. 13. It should be noted that the numerical simulations are performed under the condition of a deviation angle of 0° which represents absolutely full wake situation.

In this figure, it is obvious that the 2D_k Jensen model coupled with the RSS model yields improved predictions of wind farm power compared with the other wakes combinations models, and the results exhibit a good match with the measured data. Despite the remarkable performance of the 2D_k Jensen model, it can be seen from Fig. 13 that the predicted power reaches more or less constant after an initial drop for the second turbine, while the observed data generally decreases for the first four wind turbines and then tends to gradually increase further downwind until the last turbine. That is to say, that the 2D_k Jensen model under-predicts the wake velocity and the related power output for the last turbines. This departure is partly due to the fact that different inflow conditions are used, more specifically, the numerical predictions are carried out by assuming the deviation angle of 0° while the observed data are correspond to a wind direction sector $[-1^\circ, 1^\circ]$. In addition to this, the authors in Refs. [6,27] pointed out that the wind direction uncertainty in the dataset may leads to the discrepancy between the numerical predictions and the experimental data. For a narrowest wind direction sensor, there are only a limited number of available measurements for data analysis, which mainly attributes to the uncertainty of the dataset.

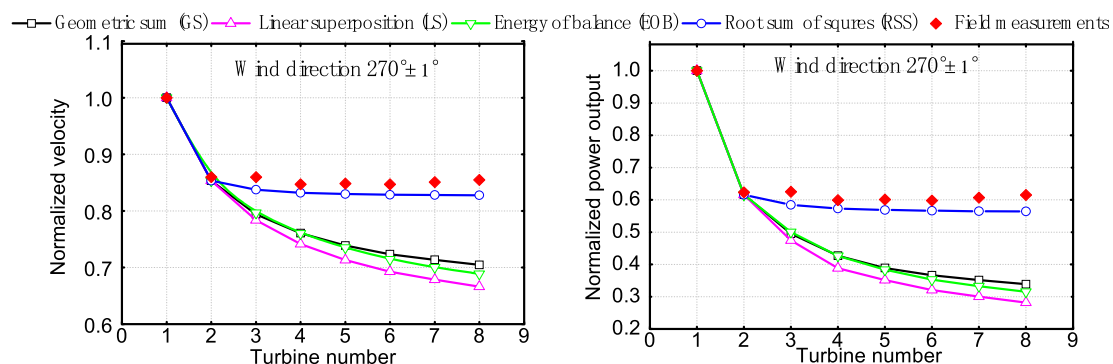


Fig. 13. Comparison of normalized wake velocity and related power output data obtained from the observations and simulations with the 2D_K Jensen model combined with 4 multiple wakes combination models for the wind sector 270° $\pm 1^\circ$.

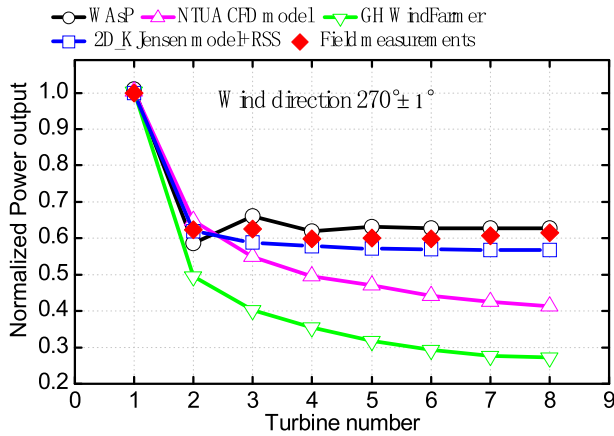


Fig. 14. Evaluation of the 2D_k Jensen wake model through comparison with other numerical methods as well as measured data.

From the comparisons of the results in this figure with those in Fig. 10 for the wind direction of $221^\circ \pm 1^\circ$ with a separation distance 7.0D, it can be found that with a change of turbine separation distance from 9.4D to 7.0D, the power output from all the downstream turbines decreases substantially, leading to a maximum reduction of 12.5% in the normalized wind power output (from 71% to 62% of the first turbine power). This can be explained by the fact that the wakes have less distance (7.0D) to be recovered before they encounter the next downstream wind turbine (in other words, the relatively short turbine separation distance limits the recovery speed of the wake flow), leading to a relatively larger velocity deficit at the turbine location and in turn, a larger power deficit when compared with the 221° case.

In order to comprehensively evaluate the performance of the proposed 2D_k Jensen model, the results obtained using other approaches [27], ranging from analytical and engineering models to computational fluid dynamics (CFD) method, are presented in Fig. 14. It should be mentioned that the numerical calculation with the 2D_k Jensen model is performed under the condition that the deviation angle is 0.5° , which is the intermediate value of the wider direction interval $[-1^\circ, +1^\circ]$. From this figure, it is evident that the wind farm design tool-GH WindFarmer and the computational code NTUA CFD predict power outputs that are much smaller than the observed power, while the WAsP software slightly over-predicts the power production and the 2D_k Jensen model combined with the RSS model predicts the power extracted by the downwind wind turbines more accurately than the other models.

There is also a strong relationship between the wind direction

and the deviation angle that influences the power deficit in addition to the turbine spacing. With the purpose to investigate the influence of wind sector on power data for the wind direction of 270° , Fig. 15 presents the wake velocity and power output obtained from the simulations with the 2D_k Jensen model, as well as the observations from Ref. [27] for the wind direction interval $[-5^\circ, +5^\circ]$. As mentioned above, the numerical results are obtained with a deviation angle of 2.5° , which is the intermediate value of wind direction sector $\pm 5^\circ$. From this figure, it is clear that the results obtained by using the developed wake model combined with the RSS model compares well with measurements, and then followed by the EOB and GS models while the LS model comes the last, which is similar to their performances in other test cases. The good agreement between the numerical predictions and the measured data, as shown in Fig. 15, again provides confidence in wind farm power prediction with the developed 2D_k Jensen model.

It is important to point out that, as shown in Figs. 13 and 15, with a small change of wind direction of only 5° from 270° , the normalized power output increases nearly 14%, from the minimum value of 0.60 (this value is the relative power as compared to the first turbine) at $270^\circ \pm 1^\circ$ to a much higher value of about 0.70 at $270^\circ \pm 5^\circ$. This is due to the fact that the wind angle $270^\circ \pm 1^\circ$ approximately represents the full wake situation, from the distribution of the wake speed along the crosswind direction as shown in Fig. 9, and the velocity deficit and consequently, the power losses are greatest at the wake center. While a departure of 5° from the full wake condition, i.e., the downstream turbines are located 0.61D apart from the wake center of the upstream one, the downstream turbines are no longer exposed to the maximum wake velocity deficit, and an increase of 11.4% of the wake velocity is observed for this partial wake condition as compared with the full wake situation.

In this study, we also employ the computational results [27] obtained using two wind farm design tools, WindSim and WAsP, as well as the LES framework coupled with an actuator disc model for the turbine power comparison. Fig. 16 reveals that these two wind farm design tools significantly over-predict the power loss as compared with the measurement. The LES method generally shows a quite good agreement with the measured data except that the drop to the third turbine is larger than the observed one, see Fig. 16. It is encouraging that the 2D_k Jensen model seems capable of providing reasonable estimations of the power losses over the whole wind turbines.

5. Conclusions

The improved 2D_k Jensen analytical wake model is presented

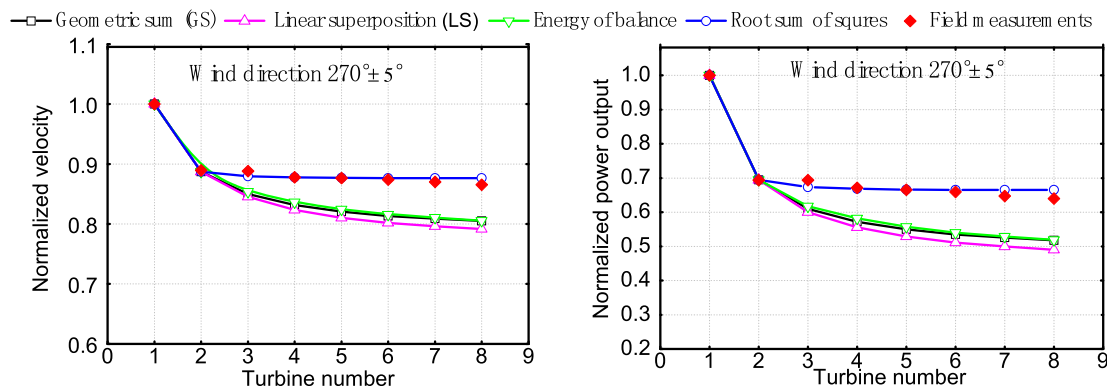


Fig. 15. Comparison of the normalized wake velocity and related power output data obtained from the observations and simulations with 2D_k Jensen model combined with different multiple wakes combination models for the wind sector $270^\circ \pm 5^\circ$.

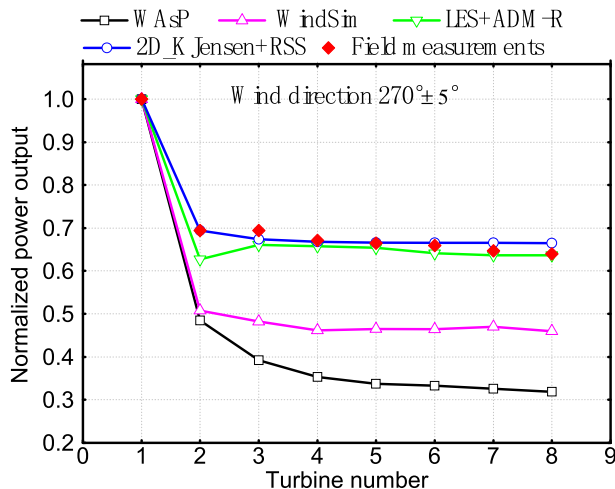


Fig. 16. Evaluation of the 2D_k Jensen wake model through comparison with other numerical methods as well as measured data.

and validated here for predicting the velocity deficit and the turbulence intensity of the wind turbine wake flow. This newly developed 2D_k Jensen wake model is based on the classical Jensen wake model and modified on the following two aspects: (1) employing a cosine shape instead of the top-hat shape to represent the distribution of the cross stream velocity in the wake; (2) employing a variable wake decay parameter derived by taking into account the ambient turbulence as well as the turbine generated turbulence. With the purpose to validate the performance of the 2D_k Jensen model when predicting the multiple wakes effect in a large wind farm, the wake interactions between two wind turbines operating in different wind conditions are simulated. The multiple wakes effect is considered through incorporating the wakes combination model into the 2D_k Jensen model, in this process four common combination models are investigated. In order to study both the full wake and partial wake operation of the downstream turbine, simulations are carried out at different stream-wise and crosswind displacements between the two rotors. The results show that the 2D_k Jensen model combined with the RSS (root sum of squares) wakes combination model predicts the wake deficit of the merging wakes at different downstream positions more accurately than coupled with the other three combination models, and provides excellent agreement with the results obtained from an advanced AL/LES model.

In addition to that, the 2D_k Jensen model coupled with the four wakes combination models are used to predict the power losses in the Horns Rev wind farm for different wake sectors around two different wind directions. These directions are 221° and 270° , corresponding to a turbine separation distance of 9.4D and 7.0D (along the prevailing wind direction) respectively. In particular, we focus on two full wake angles described above, as well as a partial wake case by adding a wind direction sector to the full wake wind angle. Hence, the effect of turbine spacing and deviation angle on the wake velocity deficit and in turn, the power output loss is investigated.

As expected, the comparisons with field measurements and other numerical methods, ranging from relatively simple engineering model to more advanced LES framework, reveal that the 2D_k Jensen wake model coupled with the RSS model exhibits a good match with the observed power data of the wind farm. Besides, as comparing the results for two different turbine spacing (9.4D and 7.0D) cases, it is observed that the power outputs of the former case are about 14% higher than that of the latter case due to

the reason that a relatively larger spacing makes the turbine wake to be recovered. When it comes to the deviation angle, it could be found that the spatial distribution of the wake velocity results in the full wake case experiencing the strongest velocity deficit and power losses while the partial wake case experiencing a relatively weaker power output deficit.

All these results consistently prove that through coupling with the multiple wakes combination model, the 2D_k Jensen model is a reliable and consistent model for predicting power production in large wind farms. Hence the present study provides a powerful model for wind farm power calculations. It is expected that the 2D_k Jensen model will be used as an optimization tool for designing large offshore wind farm arrays in our future work.

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